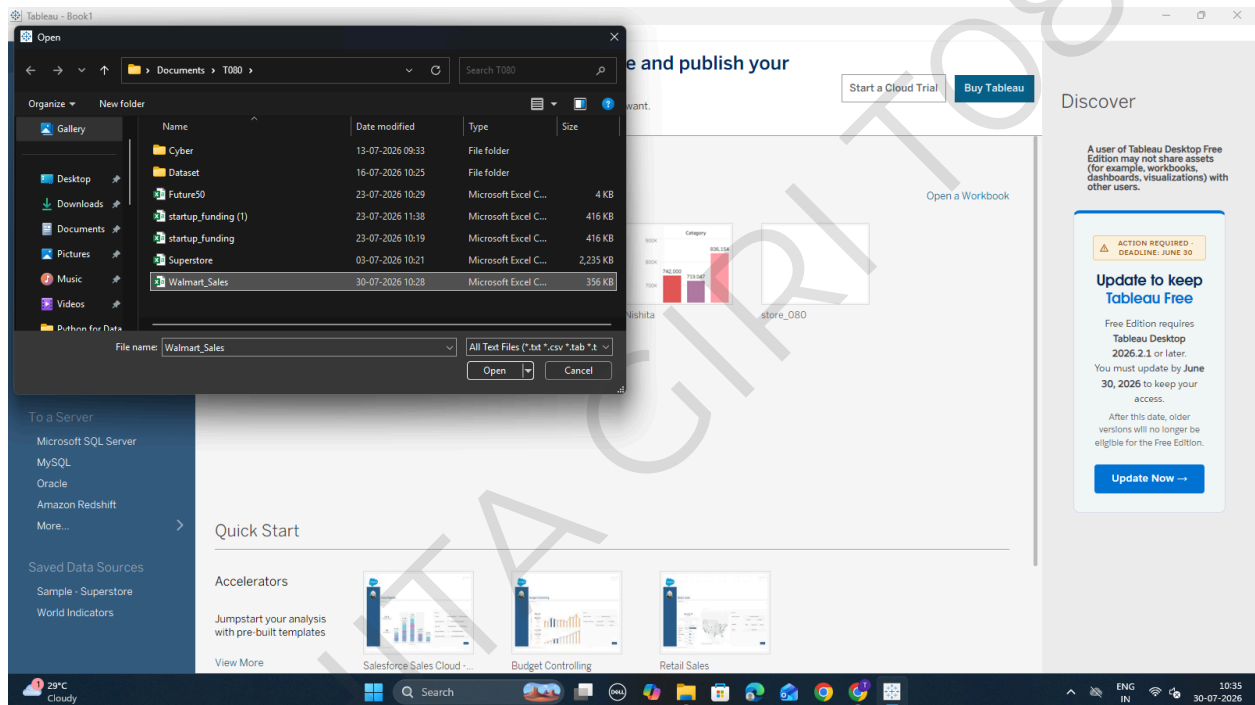


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Practical 4: Performing Data Analysis using Calculations in Tableau

- Create basic calculated fields using formulas
- Perform row-level calculations
- Perform aggregate-level calculations
- Implement FIXED Level of Detail (LOD) calculations
- Use INCLUDE and EXCLUDE LOD expressions
- Create and use parameters for dynamic analysis
- Apply calculations to fix data issues and enhance insights

a) Create basic calculated fields using formulas



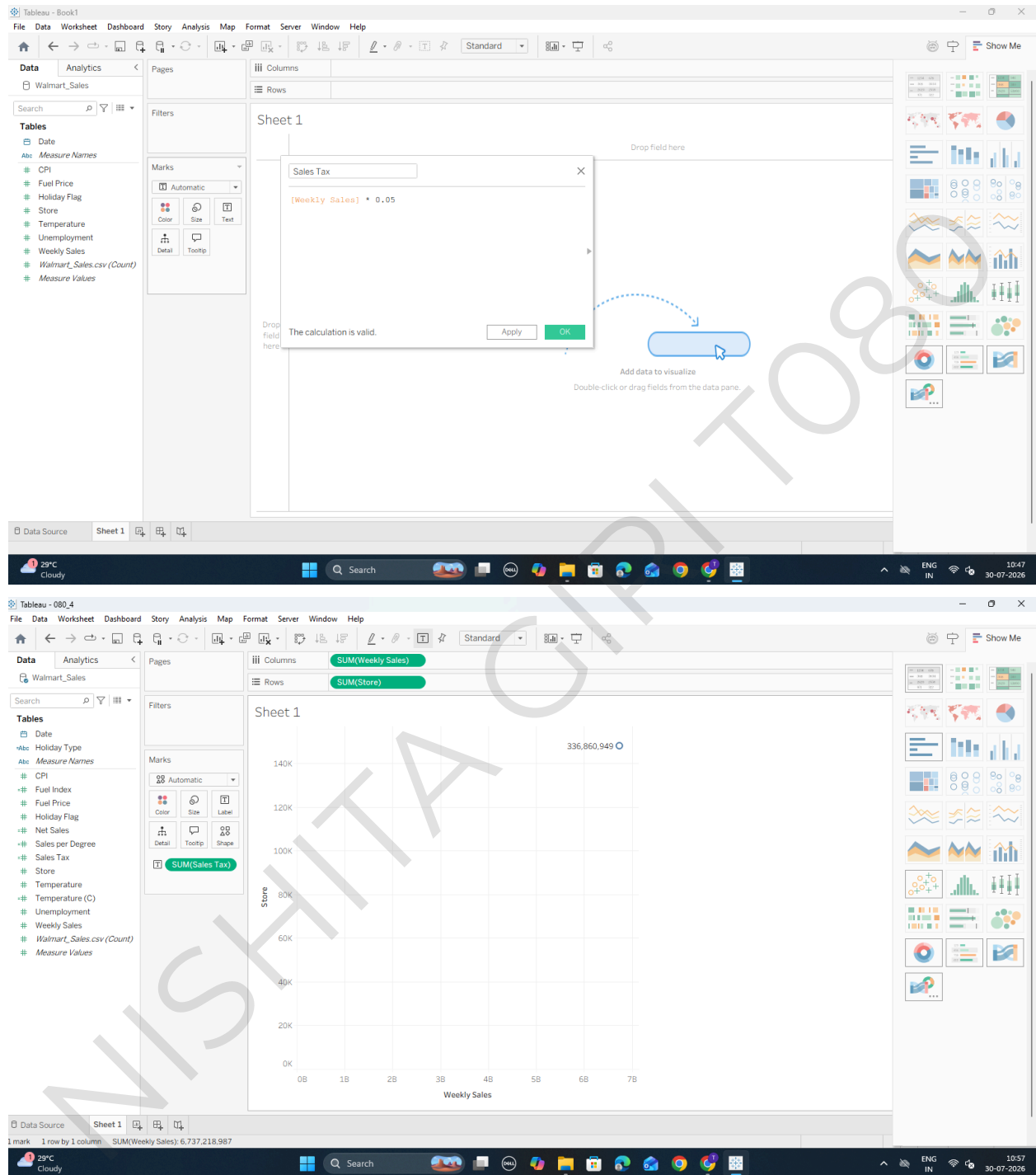
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The top screenshot displays the Tableau interface with the 'Walmart_Sales' data source selected. The table shows 8 fields and 6435 rows. The fields are: Store, Date, Weekly Sales, Holiday Flag, Temperature, Fuel Price, CPI, and Unemployment. The bottom screenshot shows the 'Create Calculated Field' menu option, which is highlighted in the 'Analytics' pane. The menu options include: Show Mark Labels, Aggregate Measures, Stack Marks, Explain Data Settings..., Reveal Hidden Data, Percentage Of, Totals, Forecast, Trend Lines, Special Values, Table Layout, Legends, Filters, Highlighters, Parameters, Create Calculated Field..., Edit Calculated Field, Infer Properties from Missing Values, Cycle Fields, Swap Rows and Columns (Ctrl+W), Analytics Extension Custom Functions Explorer, and Manage Workbook Extensions.

Store	Date	Weekly Sales	Holiday Flag	Temperature	Fuel Price	CPI	Unemployment
1	05-02-2010	1,643,690.90	0	42.3100	2.57200	211.09636	
1	12-02-2010	1,641,957.44	1	38.5100	2.54800	211.24217	
1	19-02-2010	1,611,968.17	0	39.9300	2.51400	211.28914	
1	26-02-2010	1,409,727.59	0	46.6300	2.56100	211.31964	
1	05-03-2010	1,554,806.68	0	46.5000	2.62500	211.35014	
1	12-03-2010	1,439,541.59	0	57.7900	2.66700	211.38064	
1	19-03-2010	1,472,515.79	0	54.5800	2.72000	211.21564	

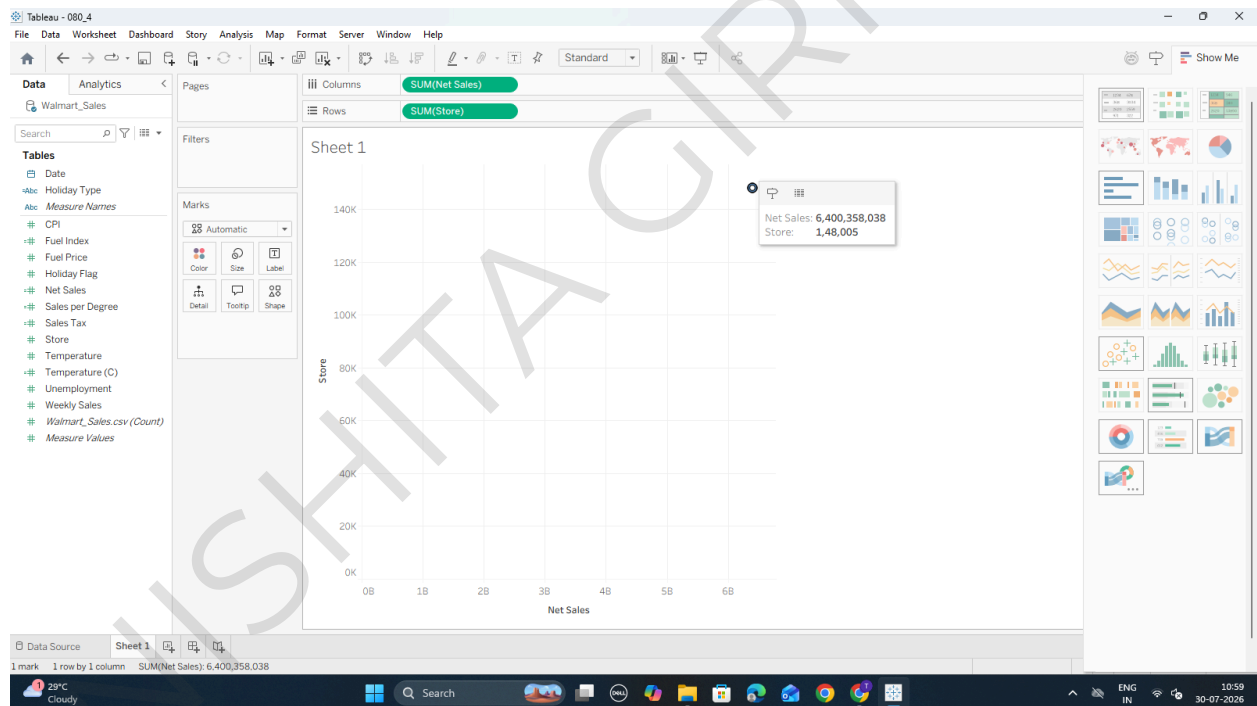
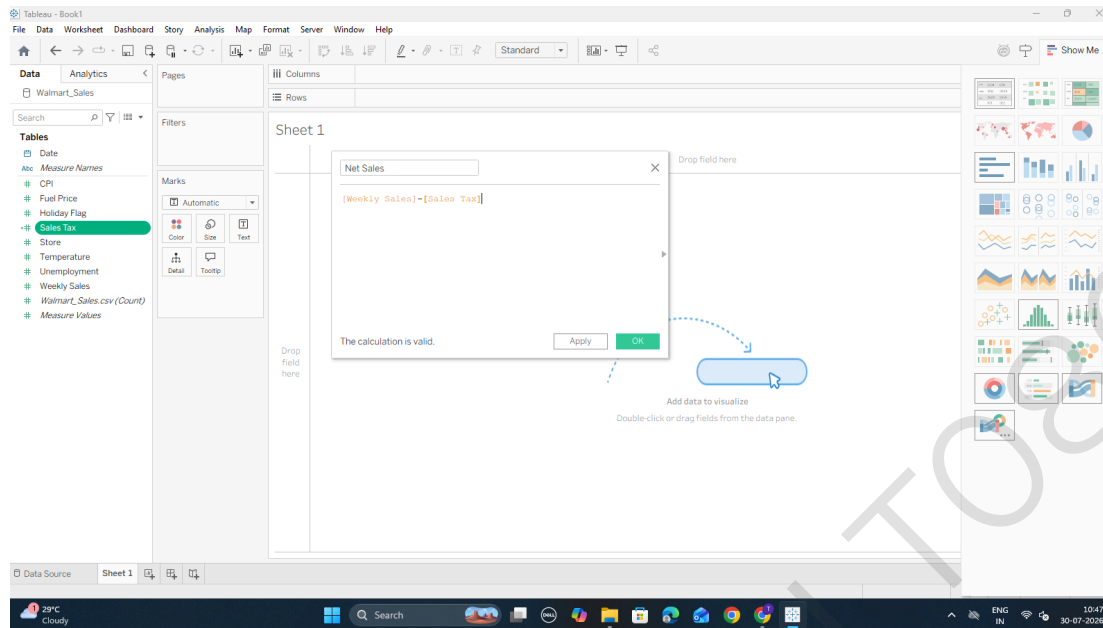
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Create Sales Tax and perform

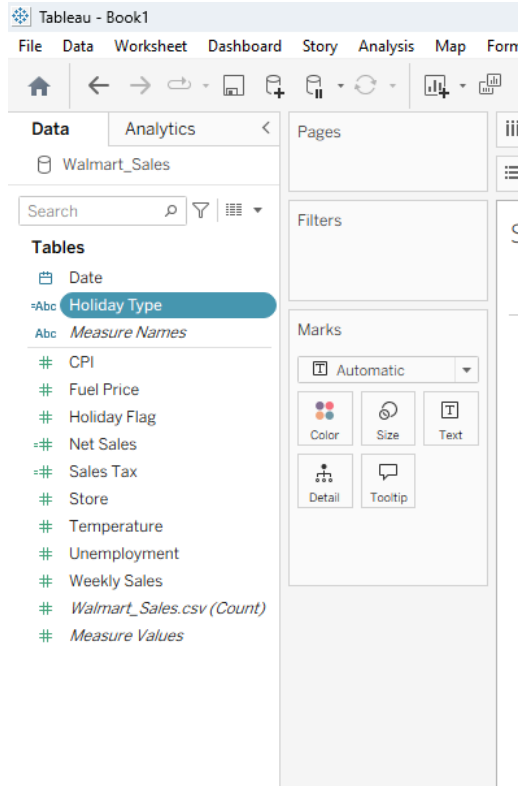


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Net Sales

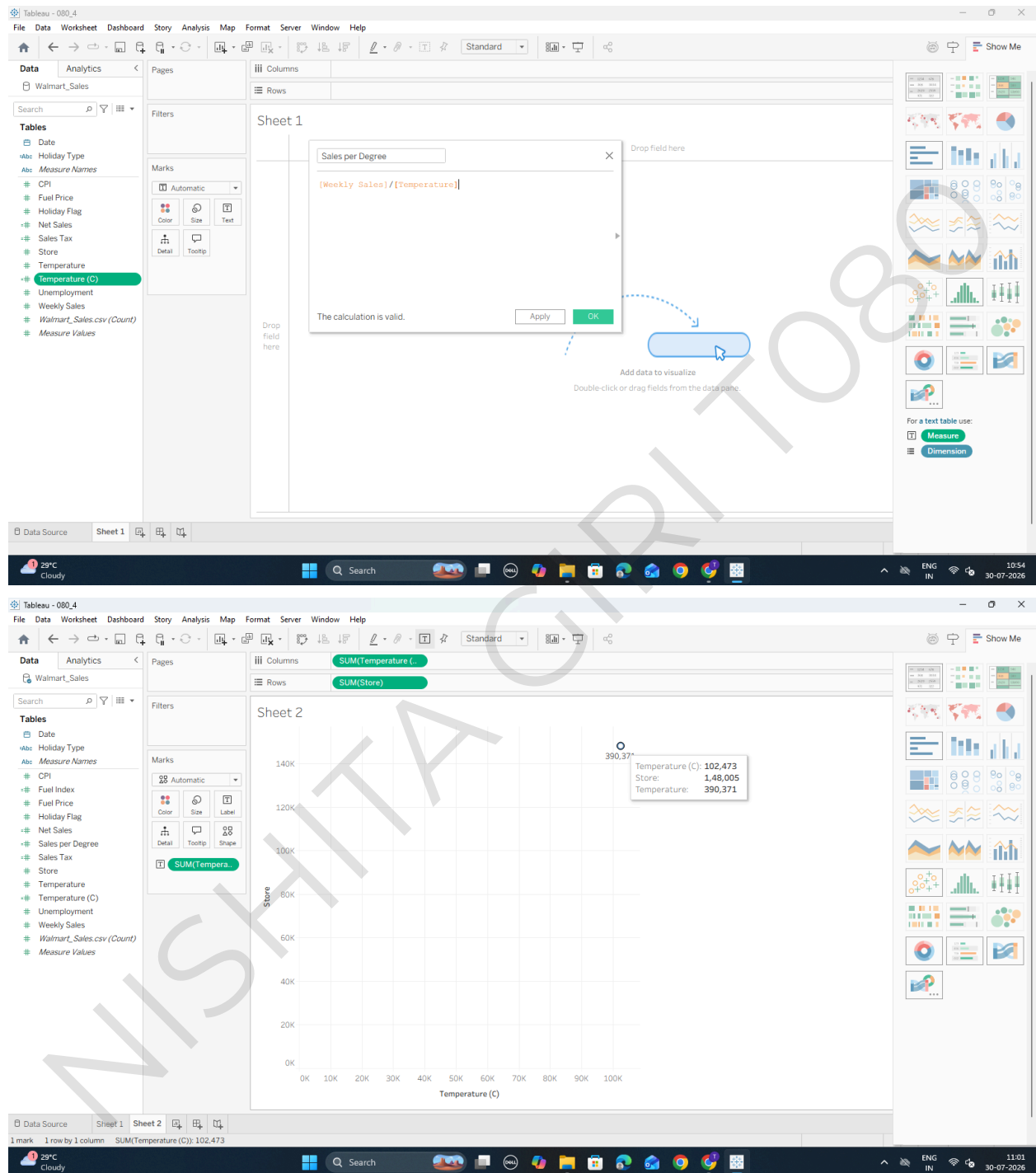


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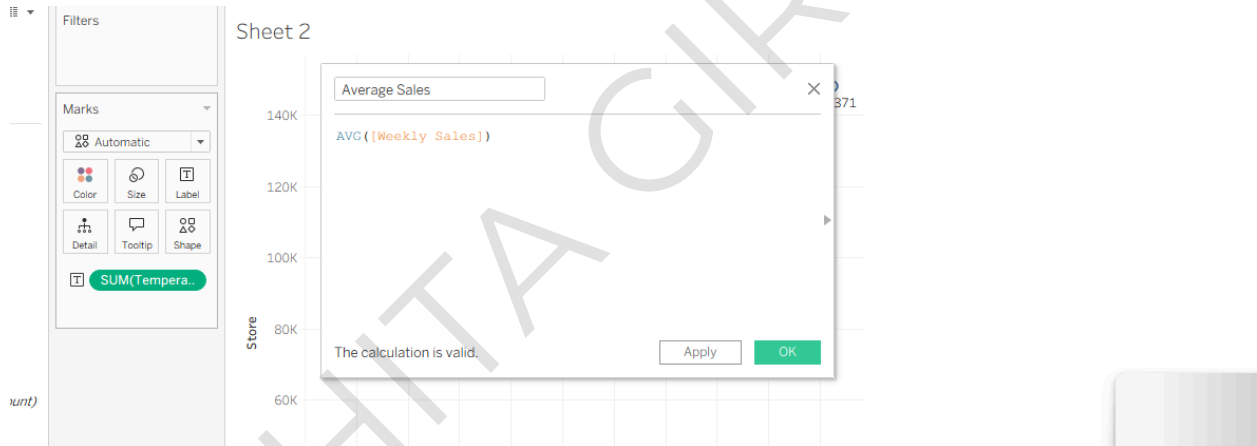
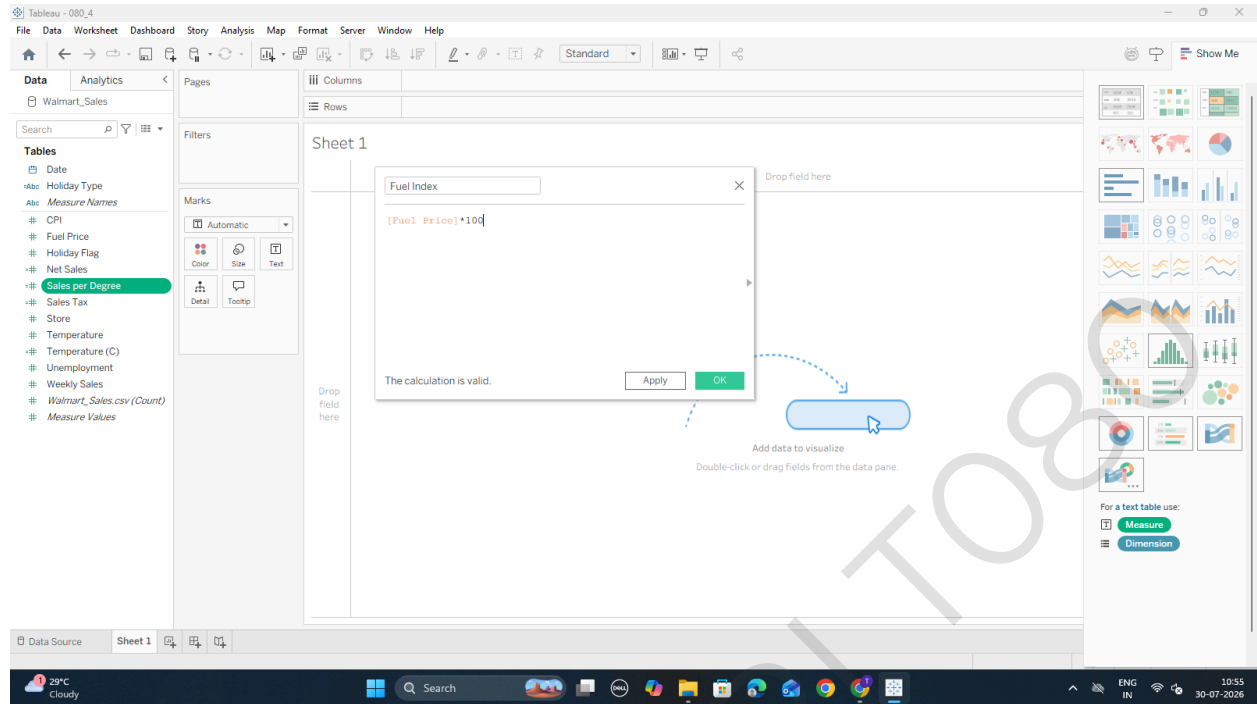


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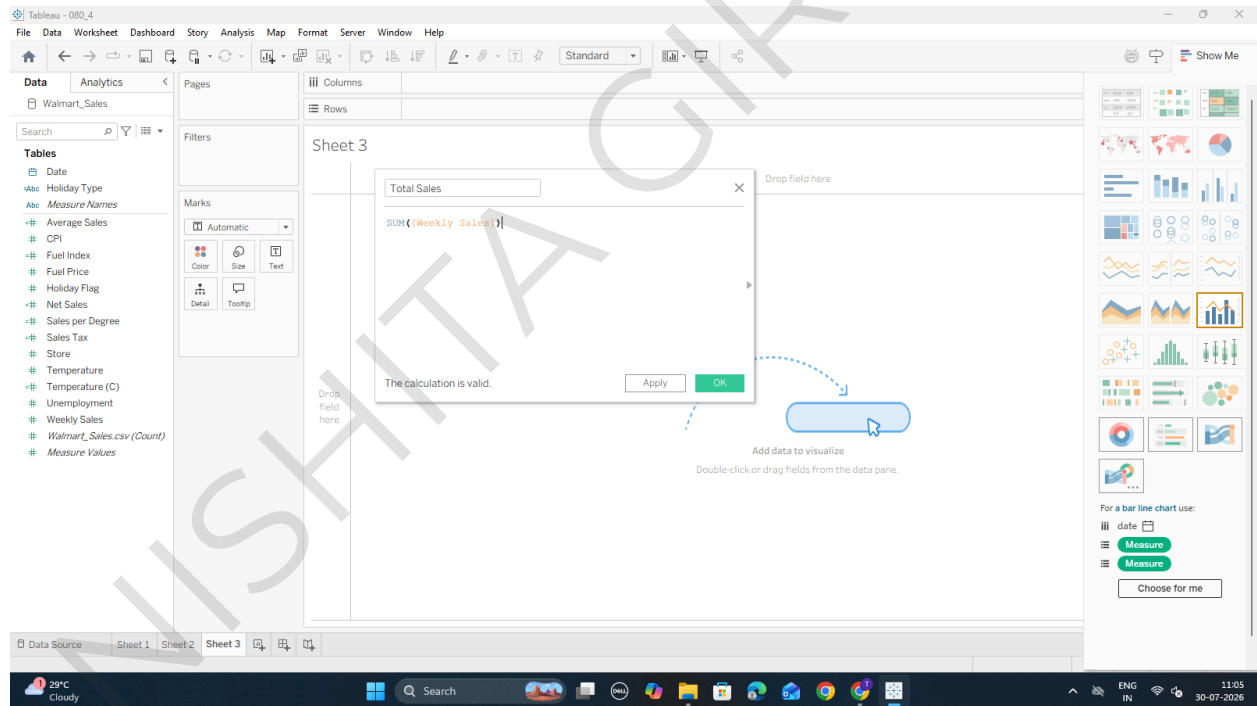
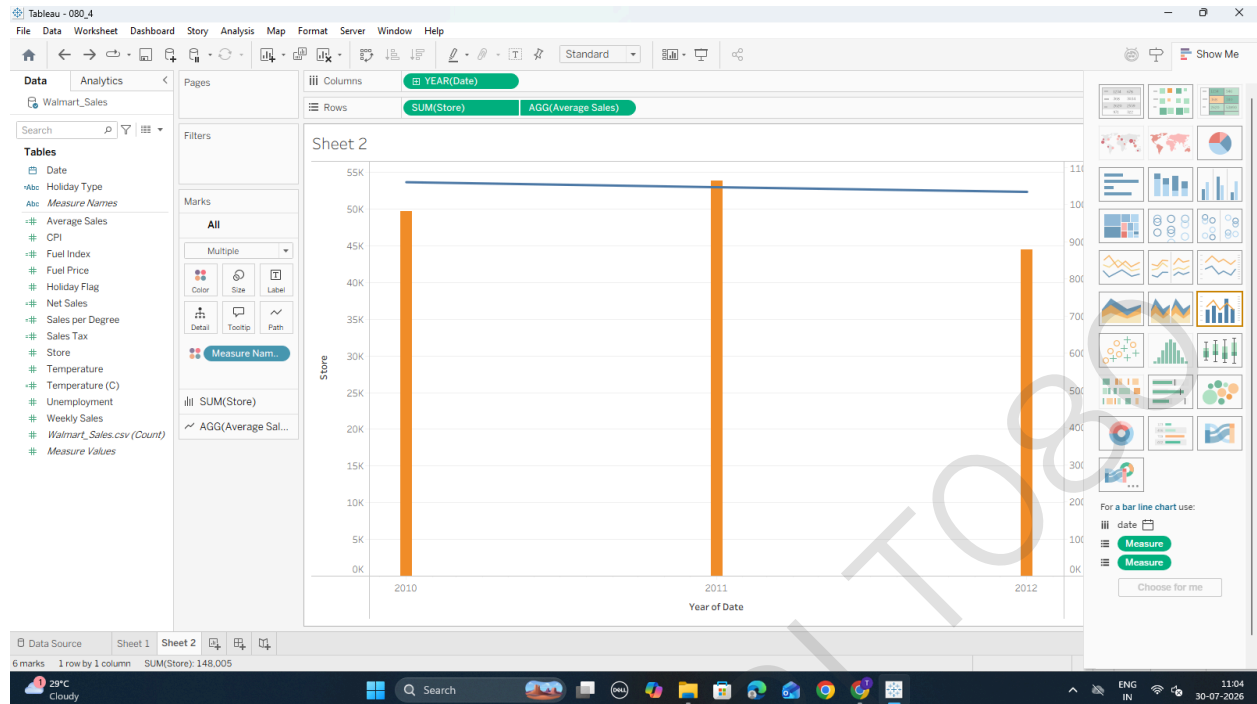
b) Perform row-level calculations



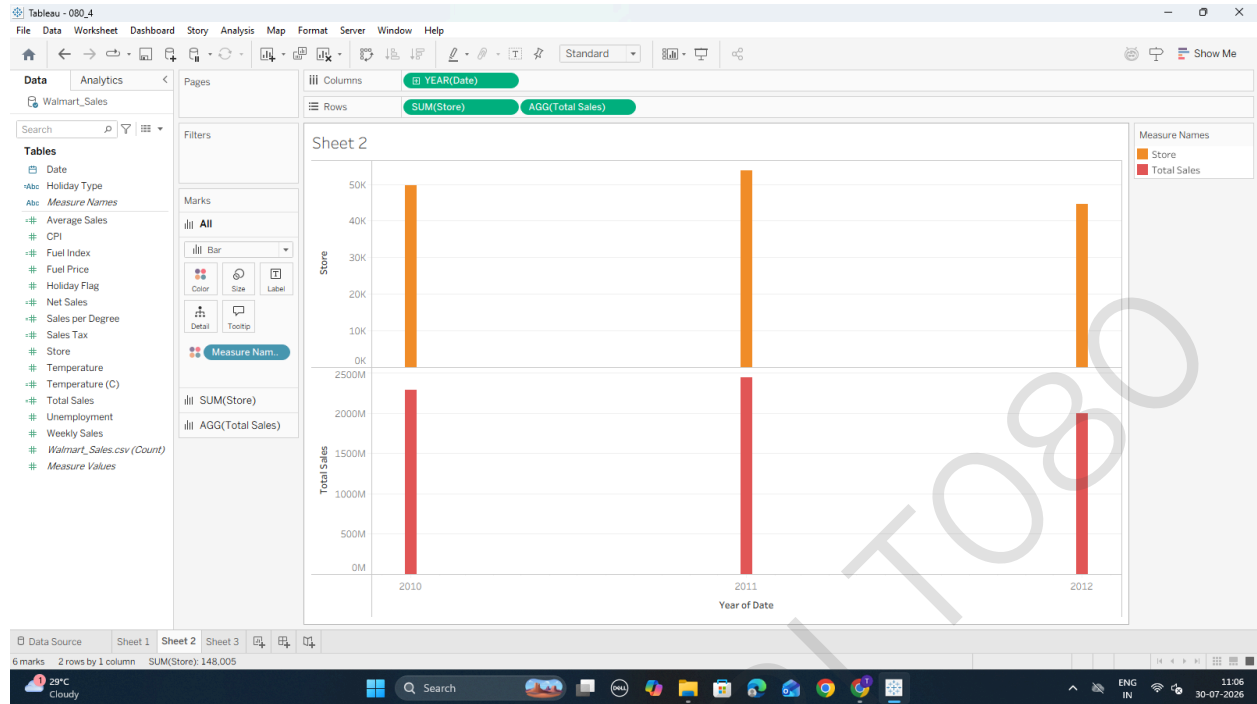
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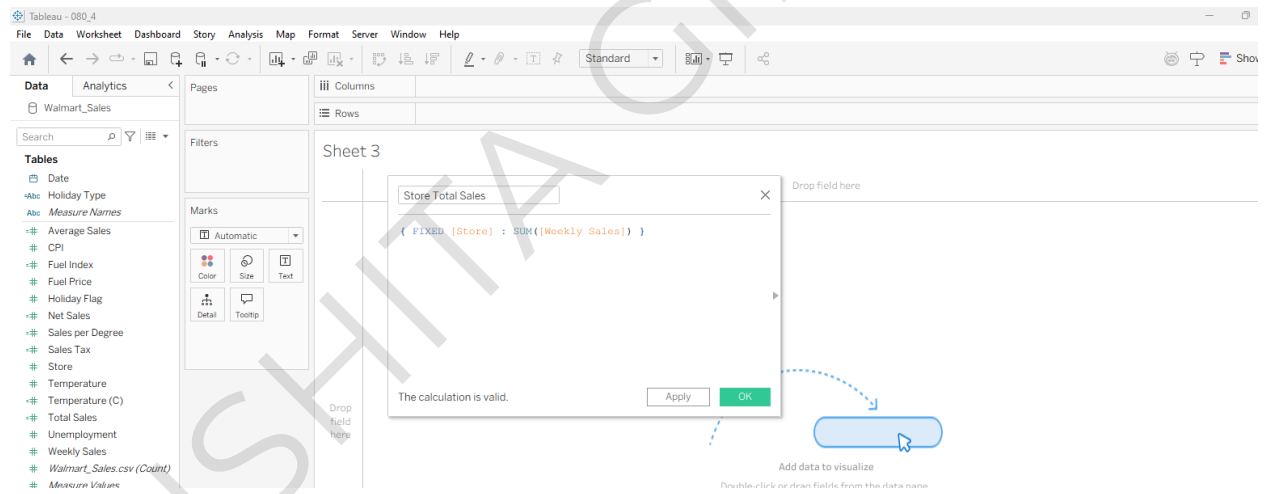
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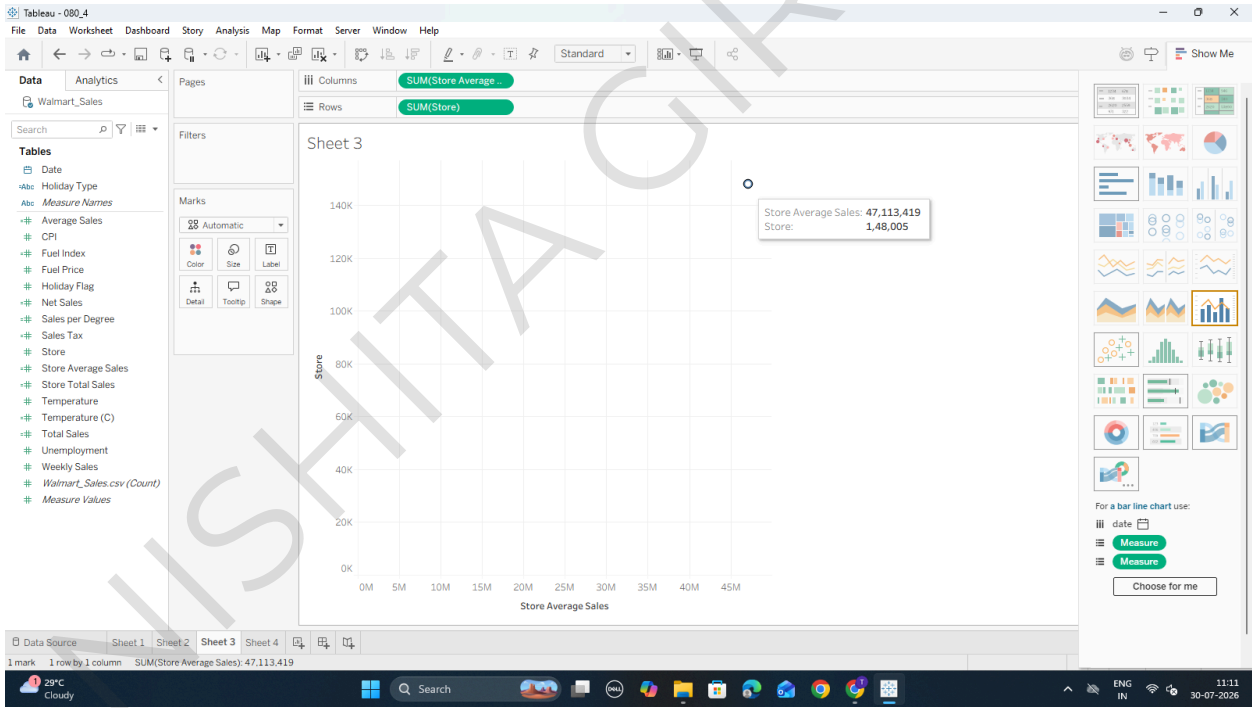
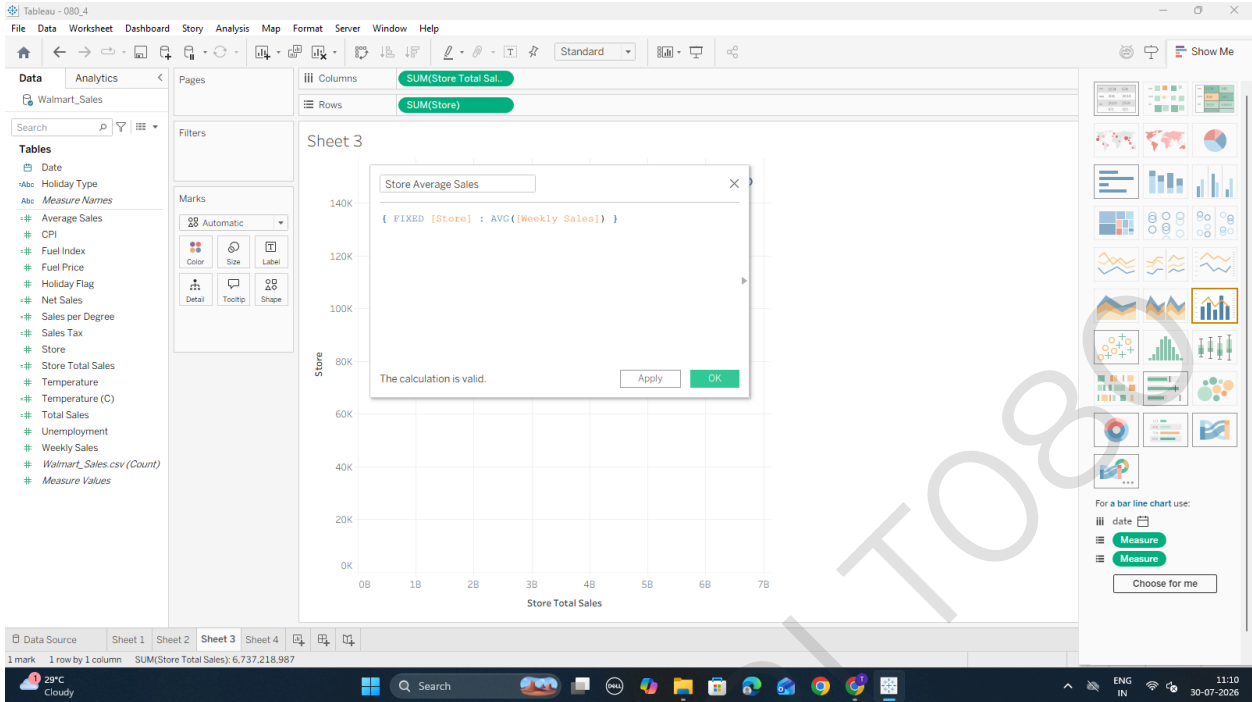
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d) Implement FIXED Level of Detail (LOD) calculations

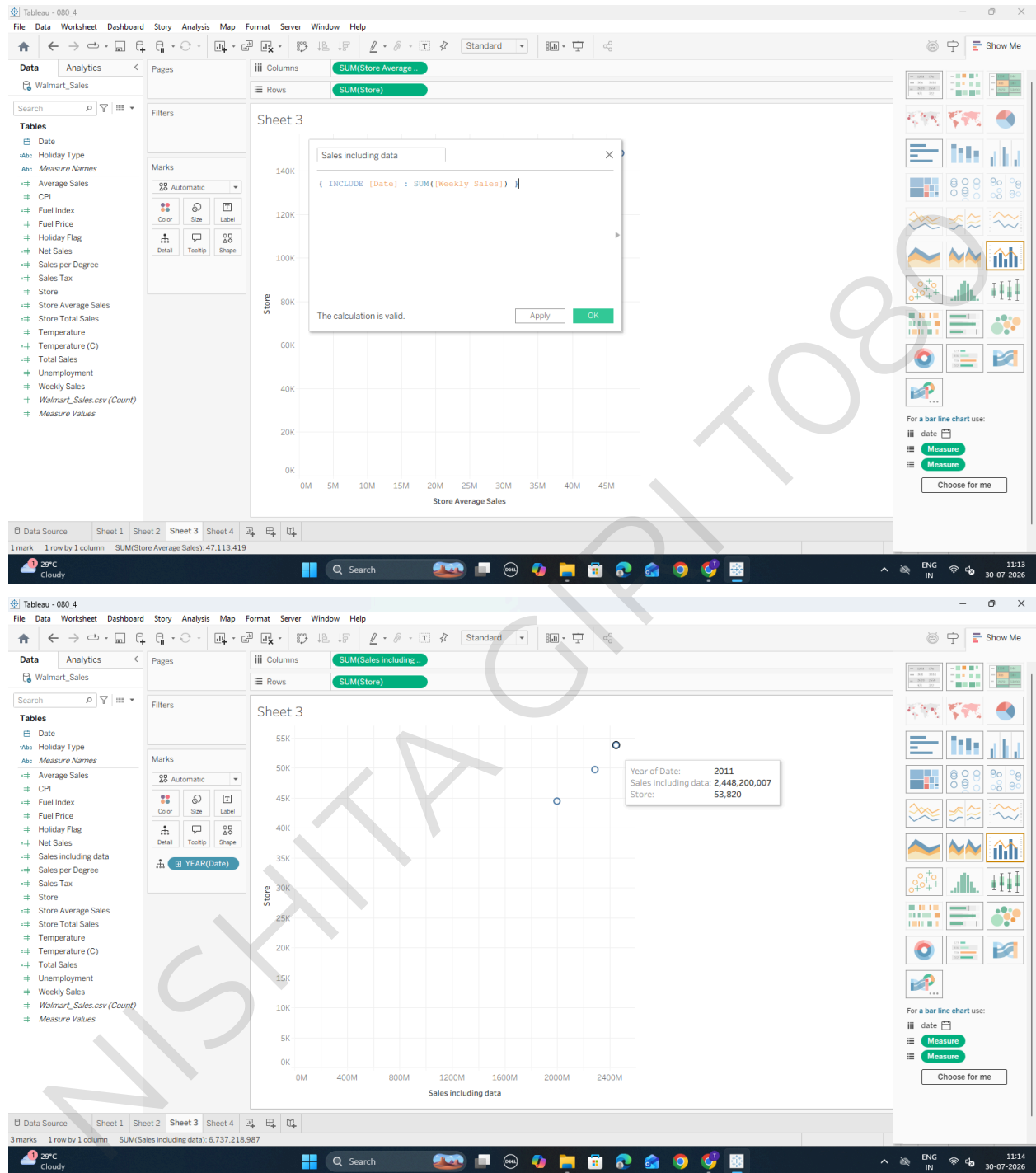


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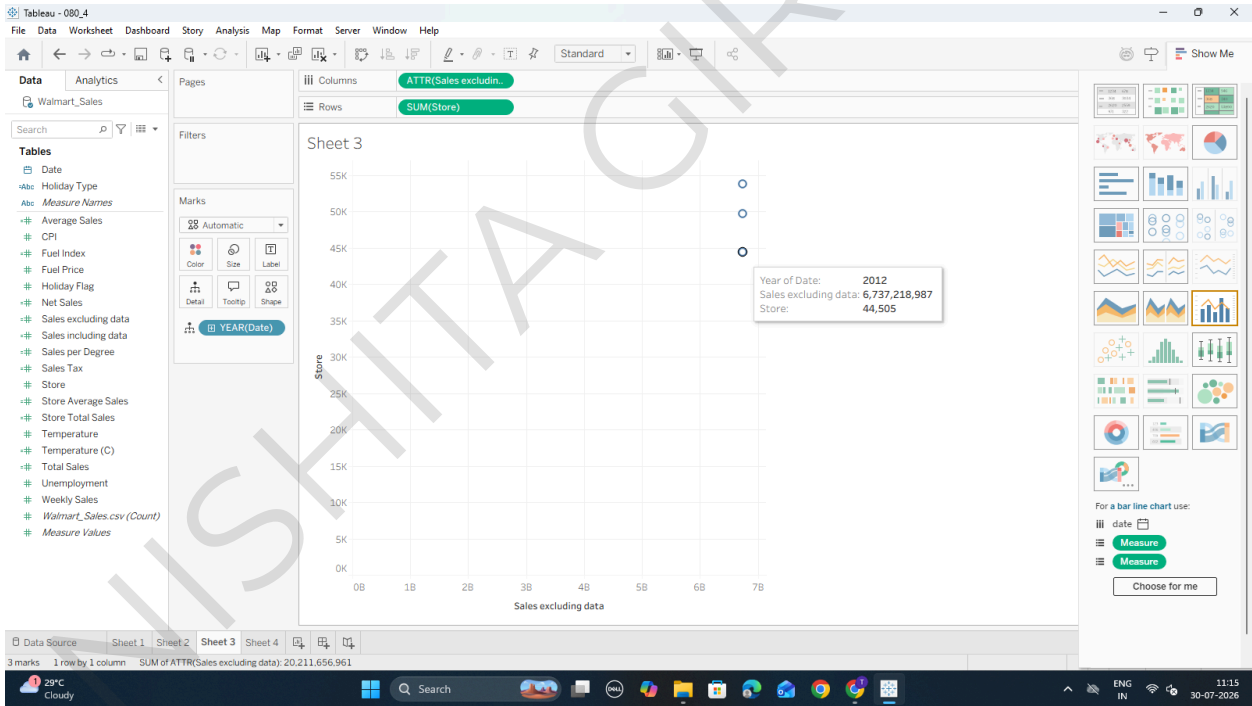
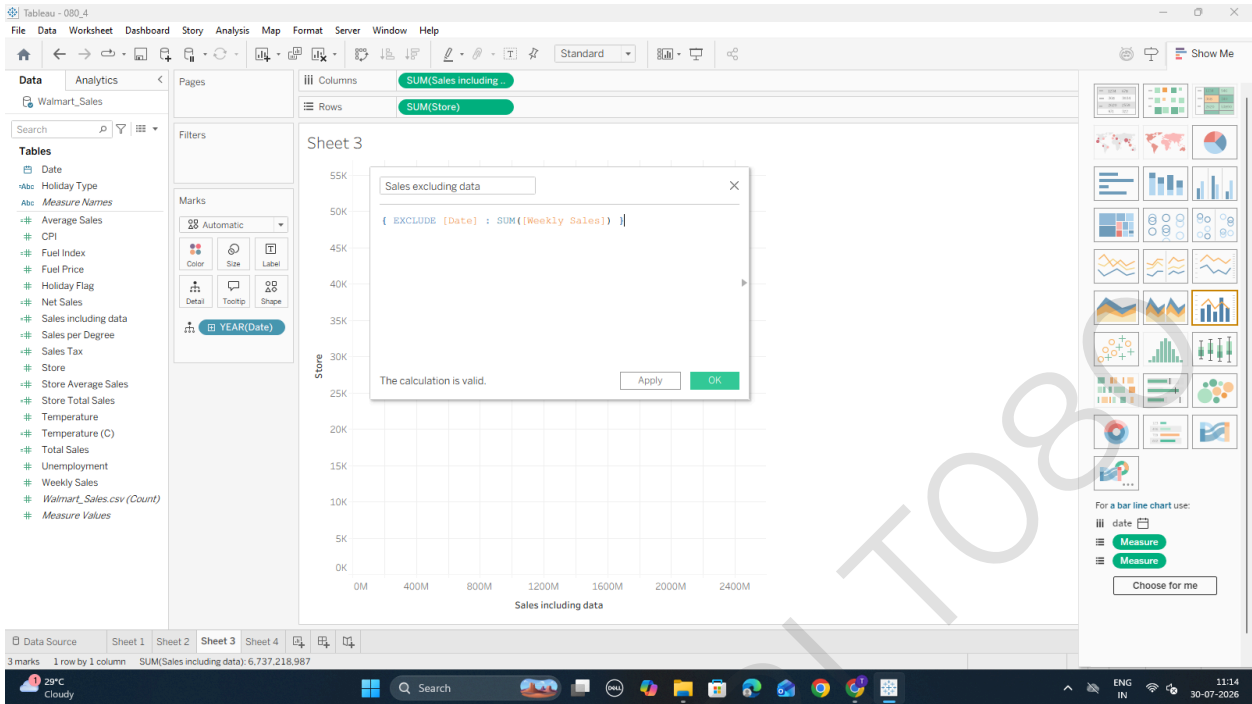


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e) Use INCLUDE and EXCLUDE LOD expressions



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f) Create and use parameters for dynamic analysis

The image displays two screenshots of the Tableau Desktop interface, illustrating the process of creating and using a parameter for dynamic analysis.

Top Screenshot: Creating a Parameter

The "Create Parameter" dialog box is open, showing the following configuration:

- Name:** Sales Threshold
- Properties:**
 - Data type:** Float
 - Display format:** 500,000
- Current value:** 500000
- Value when workbook opens:** Current value
- Allowable values:** All (selected)

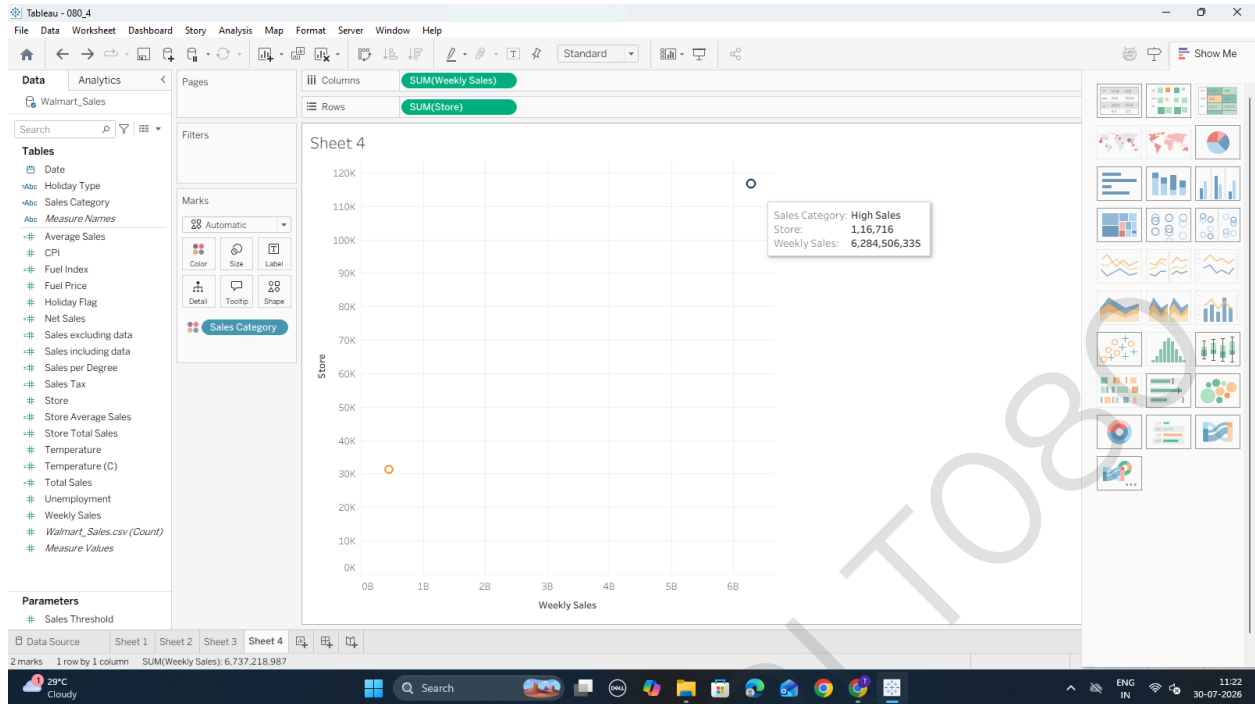
Bottom Screenshot: Using the Parameter

The "Sales Category" calculation is shown in the Columns shelf:

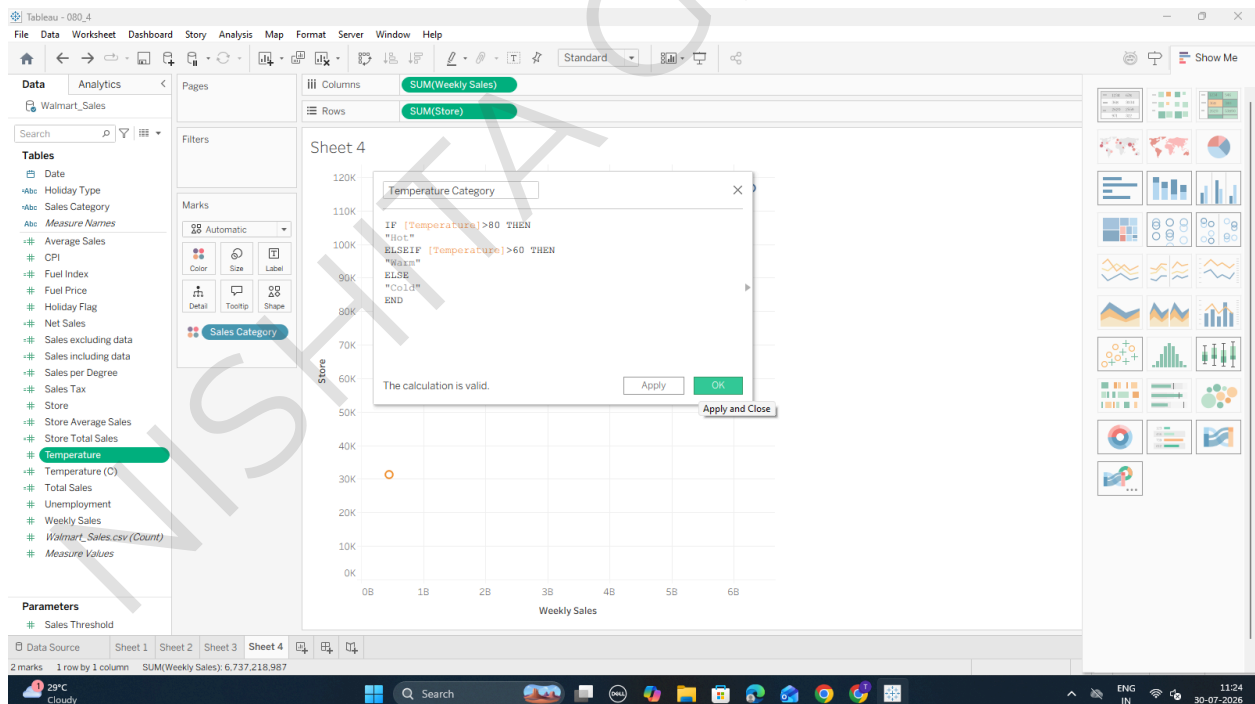
```
IF [Weekly Sales] > [Sales Threshold] THEN  
    "High Sales"  
ELSE  
    "Low Sales"  
END
```

The "Parameters" pane on the left shows the "Sales Threshold" parameter. A blue arrow points from the parameter name in the pane to the calculation in the Columns shelf, indicating its use in the formula.

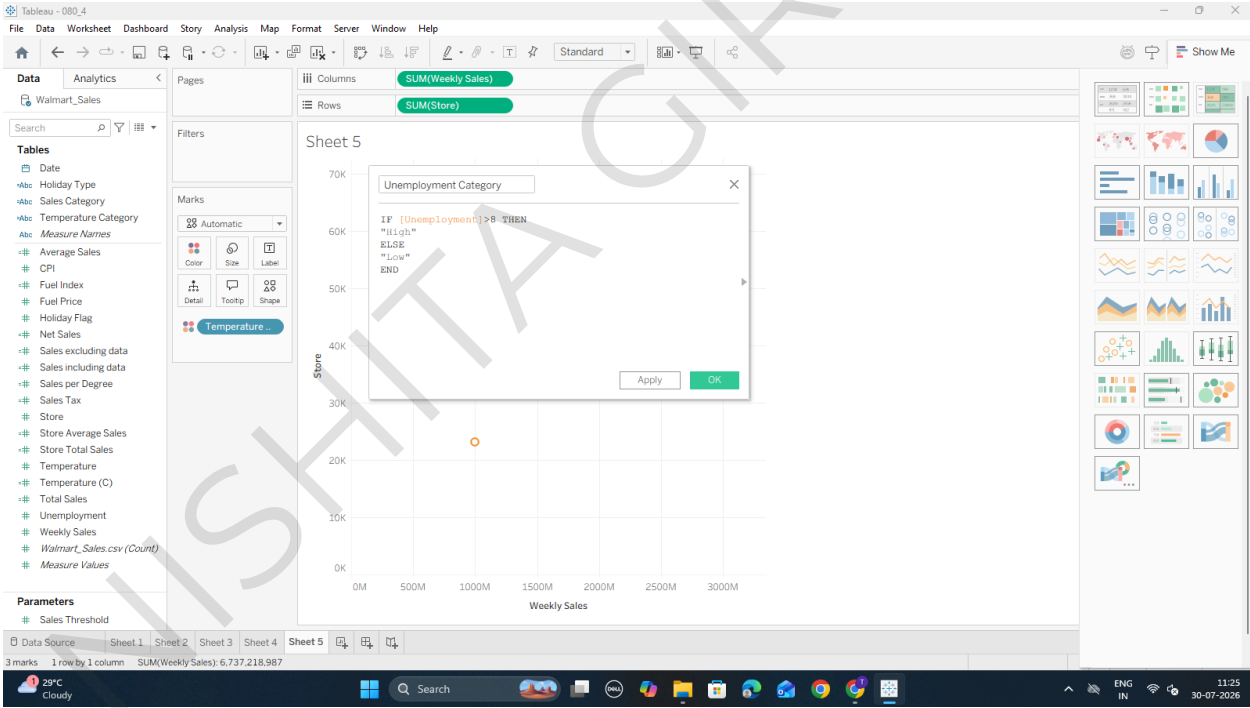
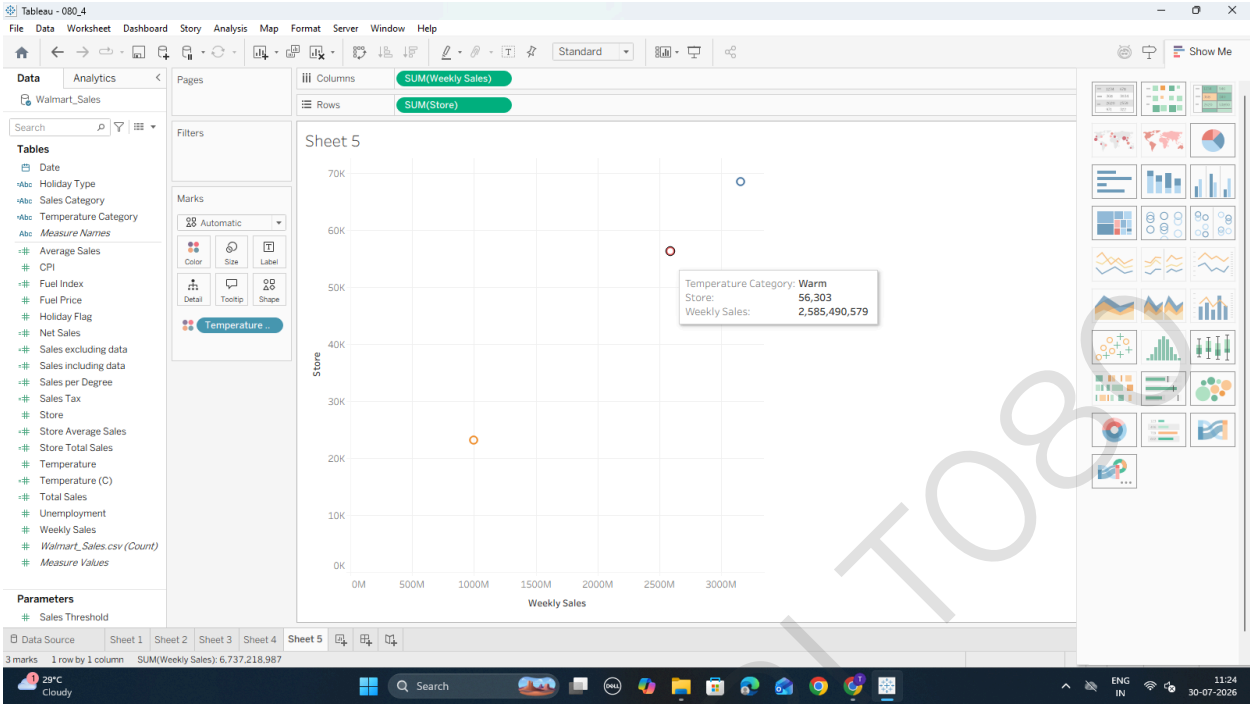
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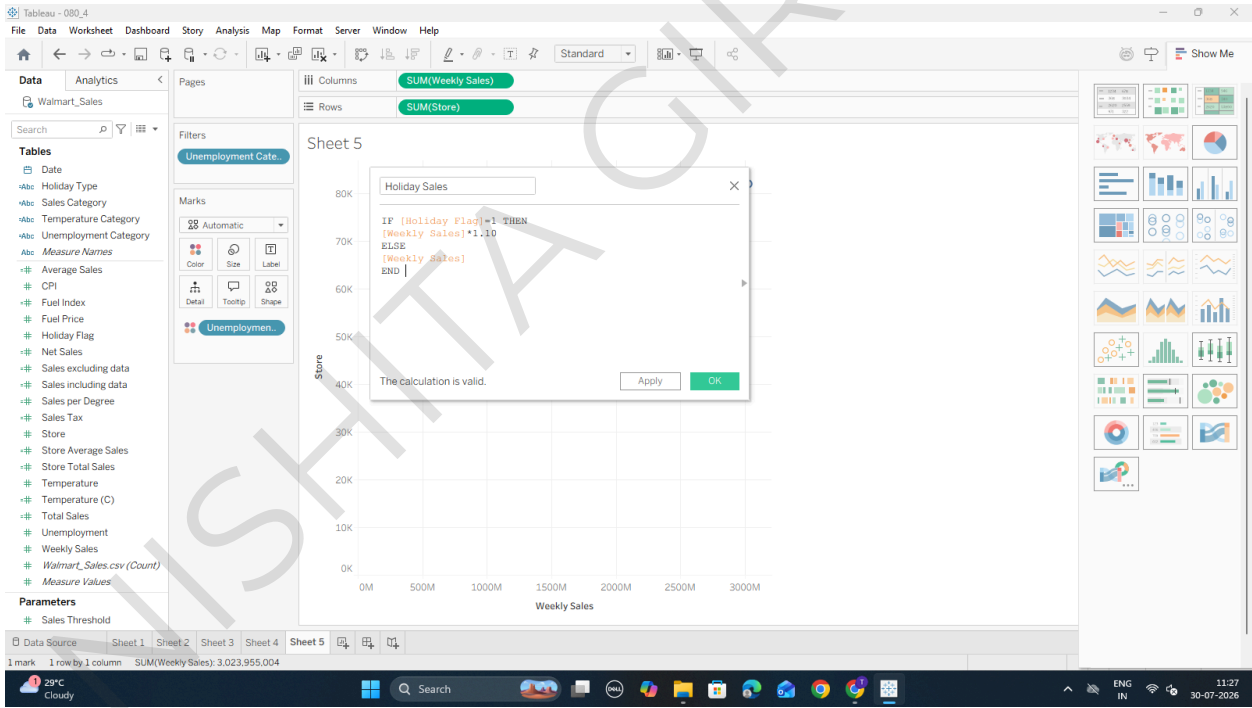
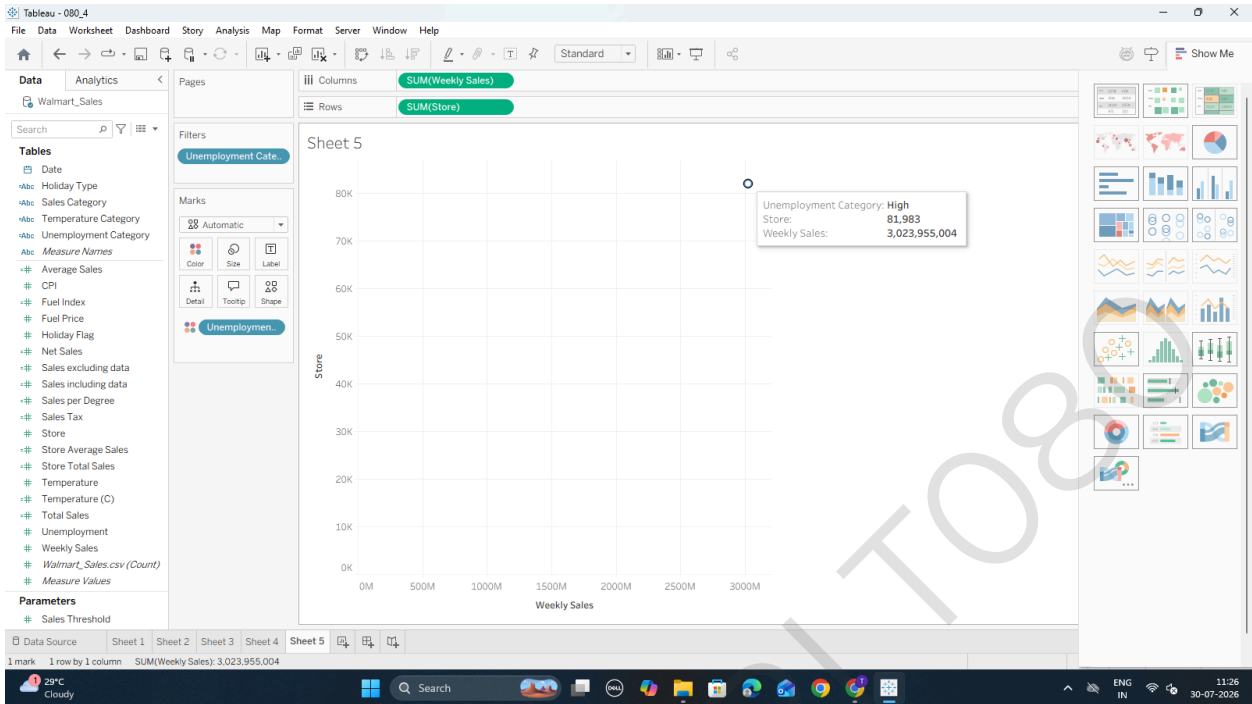
g) Apply calculations to fix data issues and enhance insights



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